

Question 1 [2p]

Fill in the blanks : Facilities design can be summarized as creating __PLACES/SPACES_ that maximizes __EFFICIENCY__

Question 2 [8p]

Name the 4 major Facility Layout types and define for what volume/variety manufacturing situations each one is ideal for.

Fixed location (Project) layout is ideal for {low} volume, {low} variety facilities - the big stuff. 2p

A *Work center (Jobshop)* layout is ideal for {low} volume, {high} variety type work. 2p

Assembly lines (Process) are ideal for {high} volume, {low} variety products. 2p

Cellular (Group technology) layouts work well for {medium} volume and {medium} variety type products. 2p

Question 3 [10p]

A manufacturer of electrical products needs to assemble 1000 units in an effective 7½ hour working day. Given the assembly tasks, standard time and preceding tasks for each,

Develop the balanced workstation configuration (number of workstations and allocation of tasks) by drawing a Precedence diagram, calculating the Cycle time and determining the minimum nr of workstations, to achieve the production target. Indicate the association of tasks to Workstations clearly either on the Precedence diagram or in a table. Also determine the efficiency of the proposed solution. Clearly show calculations and assumptions.

Task	Time (seconds)	Preceding Tasks
1A	15	-
2B	24	1
3C	6	1
4D	12	2
5E	18	2
6F	7	3
7G	11	3
8H	9	4
9I	14	5
10J	7	6, 7
11K	15	8, 9
12L	10	10, 11

Answer:

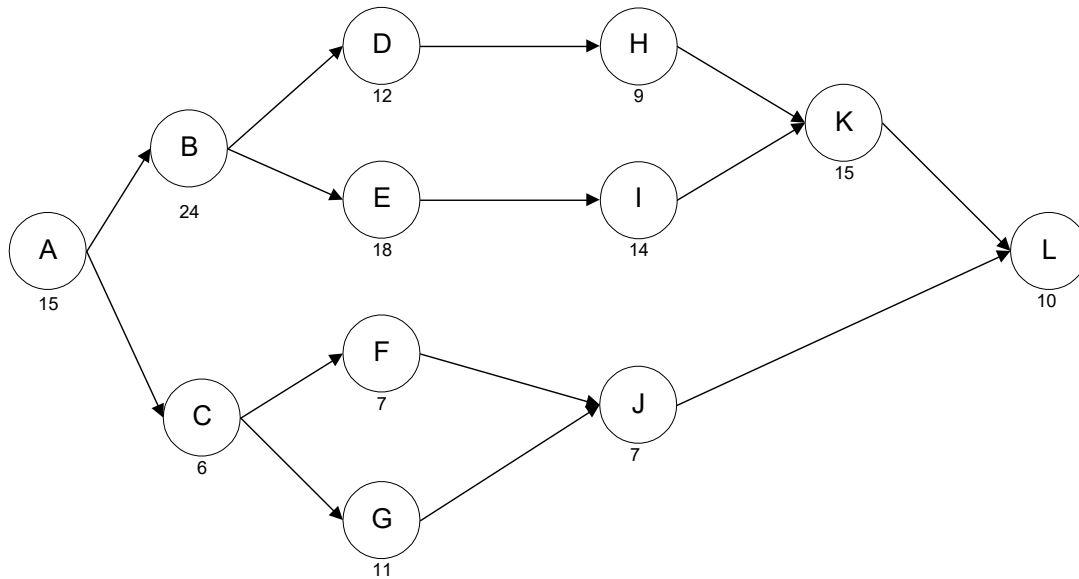
$C = \text{production time per day} / \text{required output per day}$

$$= (7.5 \text{ hour per day})(3600 \text{ seconds per hour}) / 1000 \text{ units per day} = 27 \text{ seconds per unit}$$

1p

Calculating workstations = $148\text{s} / 27\text{s} = 5.48 = 6$ workstations (minimum) 1p

1p for diagram 6p for correct task allocations per workstation



Work station	Task	Task time	Idle time
I	1 A	15	6
	3 C	6	
II	2 B	24	3
III	5 E	18	2
	6 F	7	
IV	9 I	14	1
	4 D	12	
V	7 G	11	0
	8 H	9	
	10 J	7	
VI	11 K	15	2
	12 L	10	

a. $\text{Efficiency} = \frac{T}{N_o C} = \frac{148}{6(27)} = .914 \text{ or } 91.4\%$ 1p

Question 4 [2p]

Now calculate the nr of machines needed at workstation B all doing the same task. The day is still 7,5hrs to produce 1000 units. A machine has a standard machining time of 2min/part. This type of machine is available 85% of the time due to maintenance required every now and then. During the operation time, parts are processed at 98% of the std rate.

How many machines are needed to make production demand? (remember $F = SQ/EHR$)

F = nr of machines per shift

S = std time (minutes) per unit

Q = nr of units per shift

E = actual performance as a % of std time

H = amount of time (minutes) available per machine

R = reliability of machine, as % uptime

So $F = 2(1000)/(0.85)(450)(0.98) = 5,33 = 6$ machines **2p**

Question 5 [15p]

Still referring to the Manufacturing process in Question 3 above, assume that the 3 tasks done in workstation 5, include the following: Equipment (area specification), material storage and personnel requirements needed per equipment, as given below. Determine the total space needed for workstation 5.

Data given for 16 machines:

8 x pick-and-place machines (40cmx30cm) *material storage 50square centimeters*

6 x soldering machines (work is done on tables 1mx2m per operator) *material storage 40square centimetres*

2 x testing machines (50cmx50cm) *material storage 20square centimeters*

Personnel require 1mx0,5m (1 operator per machine)

Include Supervisor desk 3sqm

Aisle space allowance of 12% (round off to whole nr)

(You may draw up a table for representation of what you did. Round off calculations to 2 decimals.)

Answer in table format

Equipment	Equipment area needed (sqm)	Material area needed (sqm)	Personnel area needed (sqm)	Total (sq1pm)
PickandPlace (8)	$0,4 \times 0,3 = 0,12 \times 8 = 0,96$ 1p Or 9600sqcm	$0,005 \times 8 = 0,04$ 1p Or 400sqcm	$0,5 \times 8 = 4$ 1p Or 40000sqcm	$0,96 + 0,004 + 4 = 5$ 1p Or 50000sqcm
Solder m/c (6)	$1 \times 2 = 2 \times 6 = 12$ 1p Or 120000sqcm	$0,004 \times 6 = 0,024$ 1p Or 240sqcm	$0,5 \times 6 = 3$ 1p Or 30000sqcm	$12 + 0,024 + 3 = 15,204$ 1p Or 150240sqcm
Testing (2)	$0,5 \times 0,5 = 0,25 \times 2 = 0,5$ 1p Or 5000sqcm	$0,002 \times 2 = 0,004$ 1p Or 40sqcm	$0,5 \times 2 = 1$ 1p Or 10000sqcm	$0,5 + 0,004 + 1 = 1,504$ 1p Or 15040sqcm
			Nett total	$21,528 + \text{desk}(3) = 24,528$ sqm 1p Or 245280sqcm
			Aisle at 12%	$2,9 = 3$ 1p Or 29434sqcm
			TOTAL AREA	27,528 sqm 1p Or 274714sqcm

Question 6 [3p]

A product is made with a market estimate of 50,000 components per week. Unfortunately, there is always scrap/rejects made as part of the process, at 3% per task. Calculate the Quantities that need to be Produced in each step so that 50 000 components are packaged to be sold at the end of a week.

Process tasks	Production Quantity Scheduled (units)	Expected Number of Good Units Produced
Assembly	$53140 / 0,97 = 54784$ 1p (or 54637)	53140
Testing	$51546 / 0,97 = 53140$ 1p (or 53045)	51546
Packaging	$50000 / 1 - 0,03 = 51546$ 1p (or 51500)	50 000

Question 7 [20p]

Four departments are to be located in a building of 600m x 1000m. The expected **personnel traffic flows** and area requirements for the departments are shown below.

You can assume “Closeness” rating values as follows:

If traffic flow is equal or more than 600 it is Absolutely Necessary (A) to be near each other

If traffic flow is equal or more than 400 it is Especially Important (E) to be near each other

If traffic flow is equal or more than 300 it is Important (I) to be near each other

If traffic flow is equal or more than 100 it is Ordinary (O) to be near each other

There are no departments that must not be next to each other.

Develop a Block Layout (indicating relative sizes per dept) using SLP. Include a Flow-between departments table and an Activity Relationship Chart as well as a Space Relationship Diagram with appropriate lines to indicate closeness expectations. [20p]

	A	B	C	D
A	0	250	25	240
B	125	0	400	335
C	100	0	0	225
D	125	285	175	0

A and D requires 200m x 200m

B requires 400m x 400m

C requires 600m x 600m

Answer: Flow Between diagram 3p Activity Relationship Chart 3p Space relationship diagram 10p and Block layout 4p

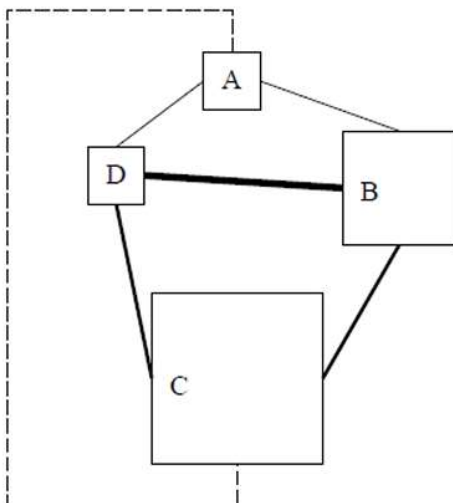
Flow-Between Diagram

	A	B	C	D
A	–	375	125	365
B		–	400	620
C			–	400
D				–

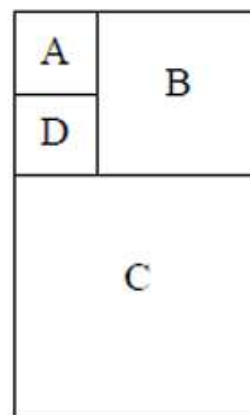
Activity Relationship Chart

	A	B	C	D
A	–	I	O	I
B		–	E	A
C			–	E
D				–

Space Relationship Diagram

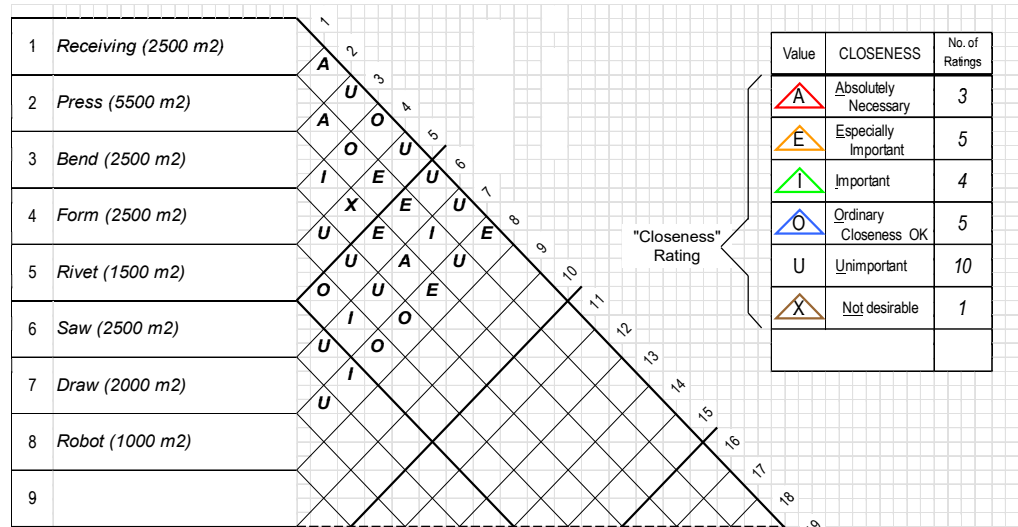


Block Layout

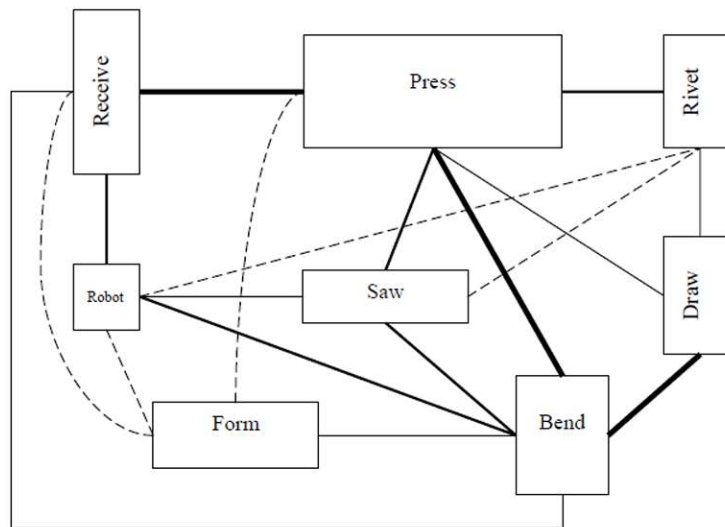


Question 8 [20p]

Use the Relationship Chart indicators below to draw up a Space Relationship Diagram as well as a Block Layout (indicating departments relative in sizes) for the 8 different departments of this facility.



17p Space relationship diagram (dept right scale and links), 3p for Layout



Block Layout

